Operating Systems Project

CPU Scheduling Simulator

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Introduction: -

This project implements a CPU Scheduling Simulator using C++. The program simulates different CPU scheduling algorithms and calculates the performance of each process. The main purpose of the project is to understand how different scheduling algorithms affect waiting time and turnaround time.

Algorithms Used: -

Algorithm	Type
FCFS	Non-Preemptive
SJF	Non-Preemptive
SJF Preemptive	Preemptive
Priority	Non-Preemptive
Priority Preemptive	Preemptive
Round Robin	Preemptive

Input and Output: -

Input:

- Number of processes
- Arrival Time
- Burst Time
- Priority

- Time Quantum for Round Robin

Output:

- Gantt Chart / Execution Timeline
- Completion Time
- Waiting Time
- Turnaround Time
- Response Time
- Average Waiting Time
- Average Turnaround Time

Scheduling Metrics: -

- Turnaround Time = Completion Time - Arrival Time
- Waiting Time = Turnaround Time - Burst Time
- Response Time = First Start Time - Arrival Time

Program Menu:

```
==================================
      CPU SCHEDULING SIMULATOR
==================================
1. FCFS
2. SJF Non-Preemptive
3. SJF Preemptive
4. Priority Non-Preemptive
5. Priority Preemptive
6. Round Robin
7. Compare All Algorithms
0. Exit
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```

Test Input:

The program was tested using four processes. Each process was given an arrival time, burst time, and priority.

```
Enter number of processes: 4

Process 1
Arrival Time: 0
Burst Time: 5
Priority: 2

Process 2
Arrival Time: 1
Burst Time: 3
Priority: 1

Process 3
Arrival Time: 2
Burst Time: 8
Priority: 3

Process 4
Arrival Time: 3
Burst Time: 2
Priority: 2
```

1. FCFS (First Come First Served)

```
Enter your choice: 1

========== FCFS ==========
Gantt Chart:
P1 (0-5)
P2 (5-8)
P3 (8-16)
P4 (16-18)

Process Arrival Burst   Priority        Completion      Waiting Turnaround      Response
P1      0       5       2               5               0       5       0
P2      1       3       1               8               4       7       4
P3      2       8       3               16              6       14      6
P4      3       2       2               18              13      15      13

Average Waiting Time = 5.75
Average Turnaround Time = 10.25
```

2. SJF Non-Preemptive (Shortest Job First)

```
Enter your choice: 2

========== SJF Non-Preemptive ==========
Gantt Chart:
P1 (0-5)
P4 (5-7)
P2 (7-10)
P3 (10-18)

Process Arrival Burst   Priority        Completion      Waiting Turnaround      Response
P1      0       5       2               5               0       5               0
P2      1       3       1               10              6       9               6
P3      2       8       3               18              8       16              8
P4      3       2       2               7               2       4               2

Average Waiting Time = 4.00
Average Turnaround Time = 8.50
```

3. SJF Preemptive (Shortest Remaining Time First)

```
Enter your choice: 3

========== SJF Preemptive ==========
Gantt Chart:
P1 (0-1)
P2 (1-2)
P2 (2-3)
P2 (3-4)
P4 (4-5)
P4 (5-6)
P1 (6-7)
P1 (7-8)
P1 (8-9)
P1 (9-10)
P3 (10-11)
P3 (11-12)
P3 (12-13)
P3 (13-14)
P3 (14-15)
P3 (15-16)
P3 (16-17)
P3 (17-18)

Process Arrival Burst   Priority        Completion      Waiting Turnaround      Response
P1      0       5       2               10              5       10              0
P2      1       3       1               4               0       3               0
P3      2       8       3               18              8       16              8
P4      3       2       2               6               1       3               1

Average Waiting Time = 3.50
Average Turnaround Time = 8.00
```

4. Priority Non-Preemptive

```
Enter your choice: 4

========== Priority Non-Preemptive ==========
Gantt Chart:
P1 (0-5)
P2 (5-8)
P4 (8-10)
P3 (10-18)

Process Arrival Burst   Priority        Completion      Waiting Turnaround      Response
P1      0       5       2               5               0       5               0
P2      1       3       1               8               4       7               4
P3      2       8       3               18              8       16              8
P4      3       2       2               10              5       7               5

Average Waiting Time = 4.25
Average Turnaround Time = 8.75
```

5. Priority Preemptive

```
Enter your choice: 5

========== Priority Preemptive ==========
Gantt Chart:
P1 (0-1)
P2 (1-2)
P2 (2-3)
P2 (3-4)
P1 (4-5)
P1 (5-6)
P1 (6-7)
P1 (7-8)
P4 (8-9)
P4 (9-10)
P3 (10-11)
P3 (11-12)
P3 (12-13)
P3 (13-14)
P3 (14-15)
P3 (15-16)
P3 (16-17)
P3 (17-18)

Process Arrival Burst   Priority        Completion      Waiting Turnaround      Response
P1      0       5       2               8               3       8               0
P2      1       3       1               4               0       3               0
P3      2       8       3               18              8       16              8
P4      3       2       2               10              5       7               5

Average Waiting Time = 4.00
Average Turnaround Time = 8.50

====================================
```

6. Round Robin

```
Enter your choice: 6

Enter Time Quantum: 2

========== Round Robin ==========
Time Quantum = 2
Gantt Chart:
P1 (0-2)
P2 (2-4)
P3 (4-6)
P1 (6-8)
P4 (8-10)
P2 (10-11)
P3 (11-13)
P1 (13-14)
P3 (14-16)
P3 (16-18)

Process Arrival Burst   Priority        Completion      Waiting Turnaround      Response
P1      0       5       2               14              9       14              0
P2      1       3       1               11              7       10              1
P3      2       8       3               18              8       16              2
P4      3       2       2               10              5       7               5

Average Waiting Time = 7.25
Average Turnaround Time = 11.75

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```

Comparison Results:

Algorithm	Avg. Waiting Time	Avg. Turnaround Time
FCFS	5.75	10.25
SJF non-preemptive	4.00	8.50
SJF Preemptive	**3.50**	**8.00**
Priority non-preemptive	4.25	8.75
Priority Preemptive	4.00	8.50
Round Robin	7.25	11.75

Result Analysis:

Based on the test results, SJF Preemptive achieved the lowest average waiting time and average turnaround time among the tested algorithms. Round Robin had the highest values in this test with a time quantum of 2.

Conclusion

In this project, I made a CPU Scheduling Simulator using C++. I implemented six scheduling algorithms and used the program to calculate the waiting time, turnaround time, and other results for each process. I also added a comparison between all the algorithms to see the difference in their performance. From the results, I noticed that the performance changes depending on the algorithm and the process of information.